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SafeRoute Project Design Document

2.1 Eisenhower Matrix and MVP

Do :

Important and Urgent

- Let users enter their starting location, destination, and planned route.
- Allow users to add family members, friends, and emergency service numbers (such as police) as emergency contacts.
- Simulate location updates to check if users are following the correct route.
- Detect when users go off route and display a warning asking if they are safe.
- Let users confirm their arrival and simulate alerts to their emergency contacts if they don't respond.
- Provide an emergency option that displays saved emergency contact information when users need help.

Schedule :

Important but Not Urgent

- Add real-time GPS tracking and an interactive map.
- Allow friends and family to follow the user's location during a trip.
- Send actual notifications to emergency contacts when something unusual happens.
- Allow the app to record locations without the Internet and sync them once the connection is restored.

Delegate :

Urgent but Not Important

- Use Java's Scanner to handle user input instead of building an input system from scratch.
- Use ArrayList to store routes and location updates.
- Use LocalDateTime and Duration to handle trip times and check-in reminders.
- Use Java's built-in file handling tools to save and load trip information.

Delete :

Not Important and Not Urgent

- Social media features for sharing trips publicly.
- A rating and review system for drivers.
- Advertisements or in-app purchases.
- A social feed where users can post about their trips.

2.2 Tech Stack

Programming Language

- Java: I'll use Java since it's the language I'm currently learning in CS 180, and I want to get more comfortable using it to build my own projects.

Libraries and Tools

- Scanner :To get information from users.

- ArrayList :To store planned routes and location updates.
- LocalDateTime and Duration :To keep track of trip times and check-in reminders.
- FileReader and FileWriter :To save and load trip information.
- Git and GitHub : To save my work and keep track of changes.

Data Storage :

- I'll use simple text or CSV files to store trip details, planned routes, and emergency contacts. I don't think I'll need a database for this first version.

External Data Sources :

- I won't need any external APIs for now since the program will use simulated locations and predefined routes. I plan to add real GPS and map services in future versions.

Development Tools :

- IntelliJ IDEA :To write and test my Java code.
- GitHub :To manage my project and track my progress.